

```
#commwnts
```

```
#nested loops
```

```
under 1 loop inside another loop
```

```
in loops we have to keep (:) colon symbol at the end
```

```
for i in range (1,5):  
    for j in range (1,5):  
        print(i,j)
```

```
1 1  
1 2  
1 3  
1 4  
2 1  
2 2  
2 3  
2 4  
3 1  
3 2  
3 3  
3 4  
4 1  
4 2  
4 3  
4 4
```

```
rows =int(input('enter the numberof rows'))
```

```
for i in range (1,rows+1):  
    for j in range (1,i+1):  
        print('*',end='')  
    print()
```

```
enter the numberof rows4  
*  
*  
*  
*  
*  
*  
*  
*  
*  
*
```

```
for i in range (1,10):  
    if i==5:  
        break  
    print(i)
```

```
1  
2  
3  
4
```

```
for i in range (1,10):  
    if i==5:  
        continue  
    print(i)
```

```
1  
2
```

```
3
4
6
7
8
9
```

```
# ths is a comment
strings=it is sequence of unicode
```

```
s="hello"
print(s)
```

```
hello
```

```
s="hello"
s=str("hello")
print(s)
```

```
"hello"
```

```
# python strings are immutable
```

```
s='hello world'
del s[0]
print(s)
```

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_928/3678770410.py in <cell line: 0>()
      2
      3 s='hello world'
----> 4 del s[0]
      5 print(s)

TypeError: 'str' object doesn't support item deletion
```

```
print('delhi'+ 'mumbai')
```

```
delhimumbai
```

```
# common features
len('aiswarya ')
```

```
10
```

```
min('aiswarya ')
```

```
' '
```

```
max('aiswarya ')
```

```
'y'
```

```
sorted('aiswarya',reverse=False)
```

```
['a', 'a', 'a', 'i', 'r', 's', 'w', 'y']
```

```
s='hello world'
s.capitalize()
s.title()
s.upper()
```

```
'HELLO WORLD'
```

```
'KeLlO wOrld'.swapcase()
```

```
'kEl1O wOrLD'
```

◆ Gemini

```
L=[1,2,3]
print(id(L[0]))
print(id[1])
print(id(1))
```

```
11645352
11645352
```

```
from types import DynamicClassAttribute
from typing_extensions import OrderedDict
#List
Ordered
Duplicated
Dynamic
ClassAttributecan remove items by slicing and indexing
NoDefaultSpecifiedhetrogenous
```

```
print([])
```

```
[]
```

```
print([1,2,3,[1,2]])
```

```
[1, 2, 3, [1, 2]]
```

```
print([1,5+6j,'true',True])
```

```
[1, (5+6j), 'true', True]
```

```
print(list('hello'))
```

```
['h', 'e', 'l', 'l', 'o']
```

```
L=[1,2,3]
L.extend([1,2])
print(L)
```

```
[1, 2, 3, 1, 2]
```

```
L=[1,2,4,5]
L.insert(2,200)
print(L)
```

```
[1, 2, 200, 4, 5]
```

```
L=[1,2,3]
L[-1]=200
print(L)
```

```
[1, 2, 200]
```

```
L1=[1,2,3,4]
L2=[1,4,5]
```

```
print(L1+L2)
print(L1*2)
print(5 in L1)
```

```
[1, 2, 3, 4, 1, 4, 5]
[1, 2, 3, 4, 1, 2, 3, 4]
False
```

```
L=[1,2,3,400]
print(len(L))
print(min(L))
print(max(L))
print(sorted(L))
L.reverse()
print(L)
L.count(2)
L.sort()
print(L)
```

```
4
1
400
[1, 2, 3, 400]
[400, 3, 2, 1]
[1, 2, 3, 400]
```

```
L=[1,2]
print(L[-1])
```

```
2
```

```
a,b,c=1,2,3
print(b,a,c)
```

```
a=(1,2,3)
b=a
a=a+(4,)
print(a)
print(b)
```

```
2 1 3
(1, 2, 3, 4)
(1, 2, 3)
```

```
s={1,2,3,4}
s.update([5,6])
print(s)
```

```
{1, 2, 3, 4, 5, 6}
```

```
L=[1,2,3]
L.remove(2)
print(L)
```

```
S={1,2,3}
S.discard(50)
print(S)
```

```
[1, 3]
{1, 2, 3}
```

```
S1={1,2,3}
S2=S1.copy()
```

```
print(S1)  
print(S2)
```

```
{1, 2, 3}  
{1, 2, 3}
```

Start coding or [generate](#) with AI.